Getting Through the Lipid Bilayer

Name _____

LT 2.5 · Read, talk, write · goes with the Membrane Transport Card Sort

Read

A wall that lets some things through

The cell membrane is a **phospholipid bilayer**: two layers of phospholipids with their water-loving heads facing the watery outside and inside of the cell, and their water-fearing tails packed together in the middle. That oily middle is what makes the membrane **selectively permeable**. Small molecules with no charge, such as oxygen and carbon dioxide, slip through it. Ions, sugars and large molecules cannot, so the cell uses proteins and vesicles to move them.

Downhill costs nothing

Particles are always moving and bumping, so they spread from where they are crowded to where they are scarce. Moving from high concentration to low is moving **down the concentration gradient**, and it takes no energy from the cell. This is **passive transport**, and it comes in three kinds. In **simple diffusion**, oxygen passes straight between the phospholipids. In **facilitated diffusion**, a channel or carrier protein gives ions and glucose a path through the oily middle; the protein opens a door but never pushes. **Osmosis** is the diffusion of water through protein channels called aquaporins. Water moves toward the side with more dissolved solute, because that side has less water. When the concentration is the same on both sides, the cell has reached **equilibrium**: particles still cross, but equally in both directions.

Uphill takes ATP

Sometimes a cell needs more of something inside than outside. A root cell gathers minerals from soil that holds very little of them, and a nerve cell keeps its sodium low. Moving particles from low concentration to high is moving **against the gradient**, and that is **active transport**. A **protein pump** uses energy from **ATP** to change shape and push particles across, one load at a time.

Cargo too big for any protein

A bacterium, a gulp of fluid or a batch of insulin is far too large to fit through a protein. The cell moves cargo like this in bulk, inside **vesicles**, and that also spends ATP. In **endocytosis**, the membrane folds inward around the cargo and pinches off a vesicle inside the cell; a white blood cell swallows a bacterium this way. In **exocytosis**, a vesicle travels to the membrane, merges with it and empties its cargo outside, the way pancreas cells release insulin into the blood. The bilayer is fluid, so it can bend, seal and merge without tearing.

Talk Take one column from your card sort. Tell your partner the picture, the name, and why that energy card fits. Then trade: which of the six was hardest to place, and what settled it?

Write Answer in full sentences. Use the bold words from the reading.

1. Oxygen crosses the membrane by simple diffusion, but a sodium ion cannot. Explain why, using the words *water-fearing tails* and *protein*.

2. Facilitated diffusion and active transport both use a protein. Explain how they differ, using *concentration gradient* and ATP.

3. A freshwater plant cell is placed in salt water. Which way does water move, out of the cell or into it? Does the cell spend ATP to move it? Explain.

4. A white blood cell takes in a whole bacterium. Name the process, and explain why the cell cannot use a channel protein.

Canvas Original #36 · © J. R. Schwebach · scienceonepagers.org · CC BY-NC 4.0